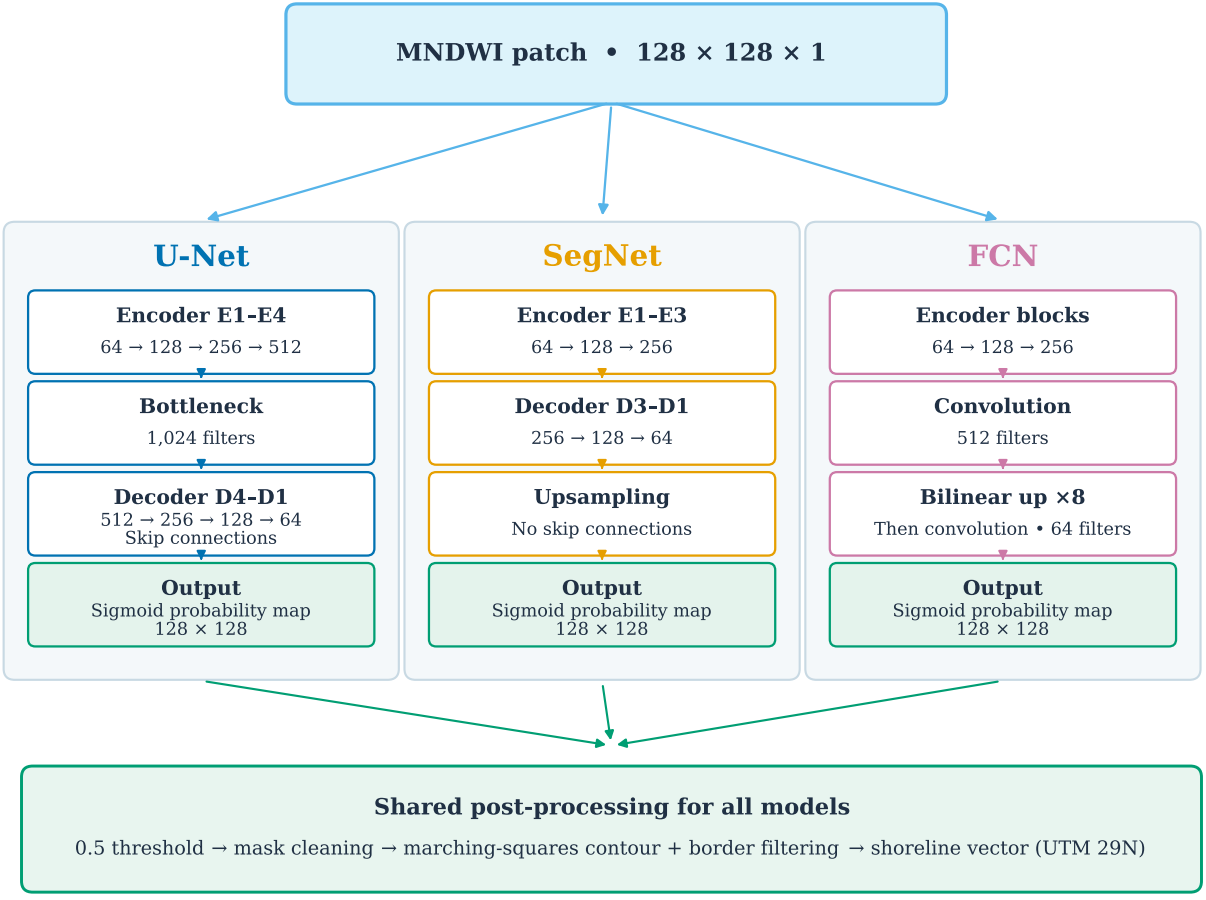
MNDWI patch • 128 × 128 × 1
U-Net
Encoder E1–E4
64 → 128 → 256 → 512
Bottleneck
1,024 filters
Decoder D4–D1
512 → 256 → 128 → 64
Skip connections
Output
Sigmoid probability map
128 × 128
SegNet
Encoder E1–E3
64 → 128 → 256
Decoder D3–D1
256 → 128 → 64
Upsampling
No skip connections
Output
Sigmoid probability map
128 × 128
FCN
Encoder blocks
64 → 128 → 256
Convolution
512 filters
Bilinear up ×8
Then convolution • 64 filters
Output
Sigmoid probability map
128 × 128
Shared post-processing for all models
0.5 threshold → mask cleaning → marching-squares contour + border filtering → shoreline vector (UTM 29N)